Add, remove, or discover LUN map reporting nodes

ONTAP 9.11.1 REST API reference

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Add, remove, or discover LUN map reporting nodes

Protocols SAN lun-maps lun.uuid igroup.uuid reporting-nodes endpoint overview

Overview

A LUN map's reporting nodes are the cluster nodes from which network paths to a mapped LUN are advertised via the SAN protocols as part of the Selective LUN Map (SLM) feature of ONTAP. SLM reduces the number of paths from the host to a mapped LUN and enables management of a single initiator group (igroup) per host.

If there are no reporting nodes in a LUN map, network paths to all cluster nodes having the appropriate network interfaces (LIFs) in the SVM are advertised. This is not a typical configuration and is reserved for limited specific use cases. Note that having no reporting nodes in a LUN map differs subtly from having all reporting nodes in the LUN map. If a LUN map has an empty reporting nodes list and a new node is added to the cluster, a path to the new node will also be advertised. If a LUN map has all cluster nodes in its reporting nodes list and a new node is added to the cluster, a path to the new node is not advertised unless the LUN map's reporting nodes are explicitly updated to include the new node.

If portsets are used to further restrict access for initiators to specific LIFs, the mapped LUN will be accessible only via the LIFs in the portset that are on the reporting nodes of the LUN map.

When a LUN map is created, the cluster node hosting the mapped LUN and its high availability (HA) partner are set as the initial reporting nodes.

Before moving a mapped LUN or a volume containing mapped LUNs to another HA pair within the same cluster, the destination node should be added to the LUN map's reporting nodes. This ensures that active, optimized LUN paths are maintained. After moving a mapped LUN or a volume containing mapped LUNs to another HA pair within the same cluster, the cluster node that previously hosted the mapped LUN should be removed from the LUN map's reporting node. Further details for this workflow may be found in the ONTAP SAN Administration documentation - see `Modifying the SLM reporting-nodes list`.

The LUN map reporting nodes REST API allows you to add, remove and discover the reporting nodes of a LUN map.

Examples

Adding a node to a LUN map

This example adds a cluster node, and its high availability (HA) partner cluster node, to a LUN map's reporting nodes.

```
# The API:
POST /api/protocols/san/lun-maps/{lun.uuid}/{igroup.uuid}/reporting-nodes

# The call:
curl -X POST 'https://<mgmt-ip>/api/protocols/san/lun-maps/c8ad35ce-ef03-
4577-88f0-9ad150ac6bbf/b10a8165-8346-11eb-ab8e-005056bbb402/reporting-
nodes' -H 'Accept: application/hal+json' -d '{ "name": "node2" }'
```

Removing a node from a LUN map

This example removes a cluster node, and its high availability (HA) partner cluster node, from a LUN map's reporting nodes.

```
# The API:
DELETE /api/protocols/san/lun-maps/{lun.uuid}/{igroup.uuid}/reporting-
nodes/{uuid}

# The call:
curl -X DELETE 'https://<mgmt-ip>/api/protocols/san/lun-maps/c8ad35ce-
ef03-4577-88f0-9ad150ac6bbf/b10a8165-8346-11eb-ab8e-
005056bbb402/reporting-nodes/6d2cd7d5-493a-daf8-9ae1-219e4ad6f77d' -H
'Accept: application/hal+json'
```

Retrieve LUN map reporting nodes

GET `/protocols/san/lun-maps/{lun.uuid}/{igroup.uuid}/reporting-nodes`

Introduced In: 9.10

Retrieves LUN map reporting nodes.

Related ONTAP commands

- `lun mapping show`

Learn more

- DOC /protocols/san/lun-maps/{lun.uuid}/{igroup.uuid}/reporting-nodes

Parameters

Name	Type	In	Required	Description
uuid	string	query	False	Filter by uuid

Name	Type	In	Required	Description
name	string	query	False	Filter by name
fields	array[string]	query	False	Specify the fields to return.
max_records	integer	query	False	Limit the number of records returned.
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned. • Default value: 1
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. • Default value: 1 • Max value: 120 • Min value: 0
order_by	array[string]	query	False	Order results by specified fields and optional [asc

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	

Name	Type	Description
num_records	integer	The number of records in the response.
records	array[lun_map_reporting_node]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "_links": {
      "node": {
        "href": "/api/resourcelink"
      },
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "igroup": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
    },
    "lun": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
    },
    "name": "node1",
    "uuid": "5ac8eb9c-4e32-dbaa-57ca-fb905976f54e"
  }
}
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
5374875	The specified LUN does not exist or is not accessible to the caller.
5374878	The specified initiator group does not exist, is not accessible to the caller, or is not in the same SVM as the specified LUN.
5374922	The specified LUN map does not exist.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
next	href	
self	href	

_links

Name	Type	Description
node	href	
self	href	

_links

Name	Type	Description
self	href	

igroup

The initiator group of the LUN map of the reporting node.

Name	Type	Description
_links	_links	
uuid	string	The unique identifier of the initiator group.

lun

The LUN of the LUN map of the reporting node.

Name	Type	Description
_links	_links	
uuid	string	The unique identifier of the LUN.

lun_map_reporting_node

A cluster node from which network paths to the LUN are advertised by ONTAP via the SAN protocols.

Name	Type	Description
_links	_links	
igroup	igroup	The initiator group of the LUN map of the reporting node.
lun	lun	The LUN of the LUN map of the reporting node.
name	string	The name of the node. Either `uuid` or `name` are required in POST.
uuid	string	The unique identifier of the node. Either `uuid` or `name` are required in POST.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Add a reporting node and its HA partner to a LUN map

POST `/protocols/san/lun-maps/{lun.uuid}/{igroup.uuid}/reporting-nodes`

Introduced In: 9.10

Adds a reporting node and its HA partner to a LUN map.

Required properties

- `node.uuid` or `node.name` - A cluster node to add.

Related ONTAP commands

- `lun mapping add-reporting-nodes`

Learn more

- DOC /protocols/san/lun-maps/{lun.uuid}/{igroup.uuid}/reporting-nodes

Parameters

Name	Type	In	Required	Description
return_records	boolean	query	False	The default is false. If set to true, the records are returned. • Default value:

Request Body

Name	Type	Description
_links	_links	
igroup	igroup	The initiator group of the LUN map of the reporting node.
lun	lun	The LUN of the LUN map of the reporting node.
name	string	The name of the node. Either `uuid` or `name` are required in POST.
uuid	string	The unique identifier of the node. Either `uuid` or `name` are required in POST.

Example request

```
{
  "_links": {
    "node": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "igroup": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
  },
  "lun": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
  },
  "name": "node1",
  "uuid": "5ac8eb9c-4e32-dbaa-57ca-fb905976f54e"
}
```

Response

```
Status: 201, Created
```

Name	Type	Description
_links	_links	
num_records	integer	The number of records in the response.
records	array[lun_map_reporting_node]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "_links": {
      "node": {
        "href": "/api/resourcelink"
      },
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "igroup": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
    },
    "lun": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
    },
    "name": "node1",
    "uuid": "5ac8eb9c-4e32-dbaa-57ca-fb905976f54e"
  }
}
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
5374875	The specified LUN does not exist or is not accessible to the caller.
5374878	The specified initiator group does not exist, is not accessible to the caller, or is not in the same SVM as the specified LUN.
5374920	The specified cluster node does not exist.
5374921	The specified cluster node name and UUID do not refer to the same cluster node.
5374922	The specified LUN map does not exist.
5374923	A cluster node `uuid` or `name` must be specified to add a reporting node.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
node	href	
self	href	

_links

Name	Type	Description
self	href	

igroup

The initiator group of the LUN map of the reporting node.

Name	Type	Description
_links	_links	
uuid	string	The unique identifier of the initiator group.

lun

The LUN of the LUN map of the reporting node.

Name	Type	Description
_links	_links	
uuid	string	The unique identifier of the LUN.

lun_map_reporting_node

A cluster node from which network paths to the LUN are advertised by ONTAP via the SAN protocols.

Name	Type	Description
_links	_links	
igroup	igroup	The initiator group of the LUN map of the reporting node.

Name	Type	Description
lun	lun	The LUN of the LUN map of the reporting node.
name	string	The name of the node. Either `uuid` or `name` are required in POST.
uuid	string	The unique identifier of the node. Either `uuid` or `name` are required in POST.

_links

Name	Type	Description
next	href	
self	href	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Remove a reporting node and its HA partner from a LUN map

DELETE `/protocols/san/lun-maps/{lun.uuid}/{igroup.uuid}/reporting-nodes/{uuid}`

Introduced In: 9.10

Removes a reporting node and its HA partner from a LUN map.

Related ONTAP commands

- `lun mapping remove-reporting-nodes`

Learn more

- DOC /protocols/san/lun-maps/{lun.uuid}/{igroup.uuid}/reporting-nodes

Response

```
Status: 200, Ok
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
5374608	An attempt was made to delete the last reporting node for a LUN. The REST API requires that a LUN map must maintain at least one reporting node.
5374875	The specified LUN does not exist or is not accessible to the caller.
5374878	The specified initiator group does not exist, is not accessible to the caller, or is not in the same SVM as the specified LUN.
5374922	The specified LUN map does not exist.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Retrieve a LUN map reporting node

GET `/protocols/san/lun-maps/{lun.uuid}/{igroup.uuid}/reporting-nodes/{uuid}`

Introduced In: 9.10

Retrieves a LUN map reporting node.

Related ONTAP commands

- `lun mapping show`

Learn more

- DOC /protocols/san/lun-maps/{lun.uuid}/{igroup.uuid}/reporting-nodes

Parameters

Name	Type	In	Required	Description
fields	array[string]	query	False	Specify the fields to return.

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
igroup	igroup	The initiator group of the LUN map of the reporting node.
lun	lun	The LUN of the LUN map of the reporting node.
name	string	The name of the node. Either `uuid` or `name` are required in POST.
uuid	string	The unique identifier of the node. Either `uuid` or `name` are required in POST.

Example response


```
{
  "_links": {
    "node": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "igroup": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
  },
  "lun": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
  },
  "name": "node1",
  "uuid": "5ac8eb9c-4e32-dbaa-57ca-fb905976f54e"
}
```


Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
5374875	The specified LUN does not exist or is not accessible to the caller.
5374878	The specified initiator group does not exist, is not accessible to the caller, or is not in the same SVM as the specified LUN.

Error Code	Description
5374922	The specified LUN map does not exist.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
node	href	
self	href	

_links

Name	Type	Description
self	href	

igroup

The initiator group of the LUN map of the reporting node.

Name	Type	Description
_links	_links	
uuid	string	The unique identifier of the initiator group.

lun

The LUN of the LUN map of the reporting node.

Name	Type	Description
_links	_links	
uuid	string	The unique identifier of the LUN.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

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